

AKHIL KOLLURI

ELECTRICAL ENGINEERING STUDENT

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EDUCATION

Iowa State University - College of Engineering - Ames, IA

B.S. Electrical Engineering | Expected May 2028

BITS Pilani - India

B.E. Electronics & Communication Engineering | 2026

TECHNICAL SKILLS

Programming: C, MATLAB

Circuit Design & Simulation: LTspice, Cadence

Digital Design: Quartus Prime, Questa

Engineering: Analog & Digital Circuits, Digital Logic, Signals & Systems, Control Systems, Electromagnetics

Design: AutoCAD

ENGINEERING LABORATORY EXPERIENCE

- Built, analyzed, and tested electronic and electrical circuits and applied circuit-analysis concepts to determine electrical quantities and evaluate experimental results.
- Documented experimental procedures, calculations, observations, and results in technical laboratory reports.
- Analyzed circuit behavior using voltage, current, resistance, and power relationships.
- Performed circuit-analysis laboratory work using node-voltage and mesh-current methods to analyze electrical networks.
- Calculated circuit quantities and evaluated circuit behavior using systematic analysis techniques.
- Completed an FPGA laboratory involving hardware design and programming using Quartus Prime and Questa.
- Worked through the design, programming, simulation, and verification process for FPGA-based digital logic.

SELECTED ACADEMIC PROJECTS AND TECHNICAL APPLICATIONS

Cadence - Circuit Design and Analysis

- Worked with electronic circuits in Cadence and applied circuit-analysis concepts to develop and analyze circuit designs.

MATLAB - Signals and Systems

- Developed MATLAB code to generate and analyze signals and created graphs to visualize signal behavior in the time and frequency domains.
- Applied signal-processing concepts through MATLAB-based calculations and graphical analysis.

LTspice - Circuit Simulation

- Built and simulated circuits in LTspice to examine circuit behavior.
- Used simulated circuit results and graphical outputs to analyze electrical quantities and circuit response.

FPGA - Quartus Prime and Questa

- Designed and programmed FPGA-based digital logic systems using Verilog.
- Simulated and verified designs using Questa and implemented designs using Quartus Prime.
- Applied digital logic and hardware-design concepts through laboratory exercises.

RELEVANT COURSEWORK

Analog Electronics & Microelectronic Circuits • Electronic Devices • Digital Design • Microprocessors & Interfacing • Signals & Systems • Control Systems • Electromagnetic Fields & Microwave Engineering • Computer Programming • Probability & Statistics

ADDITIONAL SKILLS

Technical Communication • Problem Solving • Laboratory Analysis • Engineering Documentation • Quantitative Analysis